

# STUDENT TESTIMONIAL & PROJECT SHOWCASE

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| College Name:  | V.S.M. College Of Engineering, Ramachandrapuram          |
| Branch & Year: | CSE (Data Science) / 2026                                |
| Project Title: | AI-Powered LaTeX TikZ Diagram Generator & Chatbot Studio |
| Domain:        | Education & AI Developer Software Tools                  |

## 1. Executive Summary & Internship Overview

Working on the '**AI-Powered LaTeX TikZ Diagram Generator & Chatbot Studio**' project during the IBM AICTE Internship has been a transformative, hands-on learning experience. The project focuses on solving a critical problem faced by academic researchers, professors, and engineering students: the tedious manual authoring of LaTeX TikZ diagram code.

## 2. Key Learnings & IBM Technology Integration

Through this internship project, I gained practical expertise in integrating enterprise-grade IBM AI and Cloud platforms:

- **IBM Watson Assistant WebChat:** Successfully designed and embedded an interactive conversational AI webchat agent (Integration ID: bad150a7-6816-40db-9249-10be96c432f4) hosted on IBM Cloud Sydney region (au-syd) to guide users through natural language query resolution.
- **IBM watsonx.ai & Granite LLM:** Leveraged IBM Granite models (ibm/granite-13b-instruct-v2) for multi-agent reasoning, TikZ code synthesis, and contextual code auto-correction.
- **Streamlit & Vector Compilation:** Built an interactive web application featuring real-time pdflatex and pdf2svg vector preview rendering, auto-validation, and multi-modal template galleries.

## 3. Project Impact & Real-World Value

The application bridges the gap between plain English descriptions and publication-ready academic vector graphics (Neural Networks, CNNs, Transformers, Flowcharts, Cloud Architectures). By reducing diagram creation time from hours to seconds, it provides tremendous value to the global academic and developer community.

## 4. Acknowledgements

I express my sincere gratitude to **IBM** and **AICTE** for providing this state-of-the-art virtual internship platform. The guidance, cloud infrastructure, and hands-on AI exposure have significantly boosted my technical knowledge in Data Science and Agentic AI development.

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**Student Signature:** Adapa Uma Siva Sankar

**Date:** September 2026